

API & Microservices Documentation

Internal Platform API Reference Guide

Platform Engineering Team | Version 4.1

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1. API Gateway Overview

TechViet's API infrastructure is managed through Kong API Gateway deployed on EKS. All internal and external APIs are registered through the API catalog at `api.techviet.vn`. The gateway handles authentication, rate limiting, request routing, and observability. As of March 2026, the platform serves 47 registered APIs across 12 microservices with an average daily throughput of 2.8 million requests. All APIs follow RESTful conventions documented in ADR-004 and are versioned using URL path prefixes (e.g., `/api/v2/`).

1.1 Base URLs by Environment

Environment	Base URL	Auth Method	Rate Limit	Purpose
Development	<code>https://api-dev.techviet.internal</code>	API Key	1000 req/min	Developer testing, no production data
Staging	<code>https://api-staging.techviet.internal</code>	OAuth2 + API Key	500 req/min	Pre-release integration testing
UAT	<code>https://api-uat.techviet.internal</code>	OAuth2 JWT	200 req/min	Client acceptance testing
Production	<code>https://api.techviet.vn</code>	OAuth2 JWT + mTLS	100 req/min per client	Live production traffic

2. User Service API (user-service v3.2)

The User Service manages employee profiles, authentication tokens, and role-based access control (RBAC). It serves as the identity provider for all internal applications. Deployed on 3 pods in EKS with PostgreSQL 15 backend and Redis 7 session cache. Maintainer: Le Hoang Nam (`le.nam@techviet.vn`). On-call rotation: `#team-platform` in Slack.

2.1 Endpoints

Meth od	Endpoint	Description	Auth Required	Rate Limit
GET	<code>/api/v3/users/{user_id}</code>	Retrieve user profile by ID. Returns full profile including department, role, and contact info.	Bearer Token	100/min
GET	<code>/api/v3/users?department={dept}&role;={role}</code>	Search users with filters. Supports pagination via cursor parameter. Max 50 results per page.	Bearer Token	50/min
POST	<code>/api/v3/users</code>	Create new user account. Requires admin role. Triggers onboarding workflow via Kafka event.	Admin Token	10/min
PUT	<code>/api/v3/users/{user_id}</code>	Update user profile fields. Partial updates supported. Emits <code>user.updated</code> event.	Bearer Token (self) or Admin	30/min
DELE TE	<code>/api/v3/users/{user_id}</code>	Soft-delete user account. Sets status to 'deactivated'. Revokes all active sessions.	Admin Token	5/min
POST	<code>/api/v3/auth/login</code>	Authenticate with email/password. Returns JWT access token (15min) and refresh token (7d).	None	20/min

Meth od	Endpoint	Description	Auth Required	Rate Limit
POST	/api/v3/auth/refresh	Exchange refresh token for new access token. Old refresh token is invalidated.	Refresh Token	30/min
GET	/api/v3/users/{user_id}/permissions	List all permissions and roles for a user. Includes inherited permissions from groups.	Bearer Token	50/min

2.2 Response Codes

Code	Meaning	When Returned	Retry?
200	OK	Successful GET/PUT request	N/A
201	Created	Successful POST that created a resource	N/A
400	Bad Request	Invalid request body or missing required fields	No — fix request
401	Unauthorized	Missing or expired authentication token	Yes — refresh token
403	Forbidden	Valid token but insufficient permissions for the resource	No — request access
404	Not Found	Resource does not exist or has been soft-deleted	No
409	Conflict	Email already registered or duplicate resource creation	No — use existing
429	Too Many Requests	Rate limit exceeded. Retry-After header indicates wait time.	Yes — after Retry-After
500	Internal Server Error	Unexpected server error. Incident auto-created if 5xx rate > 1%.	Yes — with exponential backoff

3. Project Service API (project-service v2.5)

Manages project portfolios, task tracking, and resource allocation. Integrates with Jira via bidirectional sync. PostgreSQL backend with full-text search via pg_trgm extension. Maintainer: Nguyen Quoc Bao (nguyen.bao@techviet.vn).

3.1 Endpoints

Meth od	Endpoint	Description	Auth	Rate Limit
GET	/api/v2/projects	List all projects. Filterable by status (active/completed/on-hold), client_id, and date range. Returns summary with health indicators.	Bearer	50/min
GET	/api/v2/projects/{project_id}	Get full project details including team members, milestones, budget utilization, and risk log.	Bearer	100/min
GET	/api/v2/projects/{project_id}/tasks	List tasks for a project. Filterable by assignee, status, priority. Supports cursor pagination.	Bearer	100/min
POST	/api/v2/projects/{project_id}/tasks	Create task within project. Auto-syncs to linked Jira board within 30 seconds.	Bearer + Project Member	30/min
GET	/api/v2/projects/{project_id}/timeline	Get Gantt chart data with milestones, dependencies, and critical path analysis.	Bearer	20/min

Meth od	Endpoint	Description	Auth	Rate Limit
GET	/api/v2/projects/{project_id}/budget	Budget breakdown: allocated vs spent vs forecasted. Includes burn rate and EAC.	Bearer + PM Role	20/min

4. Document Service API (doc-service v1.8)

Handles document storage, versioning, and full-text search. Uses MinIO (S3-compatible) for blob storage and Elasticsearch 8.x for search indexing. Supports PDF, DOCX, XLSX, PPTX, MD, and TXT formats up to 50MB per file. Documents are automatically virus-scanned via ClamAV before storage. Maintainer: Platform Team.

Meth od	Endpoint	Description	Auth	Max Size
POST	/api/v1/documents/upload	Upload document with metadata. Returns document_id and triggers async indexing pipeline.	Bearer	50 MB
GET	/api/v1/documents/{doc_id}	Download document. Returns binary stream with Content-Disposition header.	Bearer	N/A
GET	/api/v1/documents/{doc_id}/metadata	Get document metadata: filename, size, upload date, uploader, version, tags.	Bearer	N/A
GET	/api/v1/documents/search?q={query}	Full-text search across all documents. Returns highlighted snippets with relevance scores.	Bearer	N/A
DELETE	/api/v1/documents/{doc_id}	Soft-delete document. Removed from search index. Physical deletion after 90-day retention.	Admin	N/A

5. Notification Service API (notify-service v2.1)

Centralized notification system supporting email (via SendGrid), Slack (webhooks), push notifications (Firebase), and in-app notifications. Supports templated messages, batch sending, and delivery tracking. Processes approximately 15,000 notifications daily.

Meth od	Endpoint	Description
POST	/api/v2/notifications/send	Send notification. Requires channel (email/slack/push/in-app), recipient_id, template_id, and variables.
GET	/api/v2/notifications/user/{user_id}	Get notification history for user. Filterable by channel, read/unread status, date range.
PUT	/api/v2/notifications/{notif_id}/read	Mark notification as read. Batch endpoint: PUT /api/v2/notifications/read with array of IDs.
GET	/api/v2/notifications/templates	List available notification templates. Each template has variables, channels, and sample output.

6. Common Integration Patterns

All services communicate via Kafka for asynchronous events and gRPC for synchronous calls. Event schema registry is managed via Confluent Schema Registry at schema.techviet.internal. All events follow CloudEvents v1.0 specification.

Event Name	Producer	Consumers	Payload Summary
user.created	user-service	onboarding-service, notification-service, audit-log	user_id, email, department, role, created_at

Event Name	Producer	Consumers	Payload Summary
user.updated	user-service	search-service, audit-log	user_id, changed_fields, updated_by
project.status_changed	project-service	notification-service, reporting-service	project_id, old_status, new_status, changed_by
document.indexed	doc-service	search-service, chatassist-service	doc_id, filename, chunk_count, index_duration_ms
incident.created	incident-service	pagerduty-bridge, notification-service, slack-bot	incident_id, severity, title, assigned_team